6.2 The Definite Integral

AP Calculus 6.2 · Student sheet

Name _____ Period ____ Date _____

I can…

- form a Riemann sum, including from a table with unequal widths.
- write a limit of Riemann sums as a definite integral.
- evaluate a definite integral using geometry and signed area.

Vocabulary · fill in the missing word

Riemann sum — $\sum f(u_i)\Delta x_i$, where u_i is any number in the i-th _____

Definite integral — the limit of the _____ sums of f on [a, b] as the largest $\Delta x_i \to 0$

Integrand — the _____ f in $\int f(x)\,dx$

Limits of integration — the numbers a (_____) and b (upper) in $\int$ from a to b

Part 1 · Signed area

$y = x - 2$ on [0, 5].

Region	Area	Sign
Below the x-axis (0 to 2)		
Above the x-axis (2 to 5)		
Net (signed) area		

Part 2 · Notes and examples (fill in each note, then work the example)

Riemann sum: $\sum f(u_i)\Delta x_i$, where u_i is any number in the i-th _____.

Example 1

x: 0, 1, 3, 4, 7 and f(x): 2, 5, 6, 4, 3. Approximate $\int_0^7 f(x)\,dx$ with a right Riemann sum.

Definite integral: the limit of the Riemann sums as the largest $\Delta x_i \to$ _____.

Example 2

Write the limit as $n \to \infty$ of $\sum (1 + \frac{2i}{n})^2 (\frac{2}{n})$ as a definite integral.

Signed area: area above the x-axis counts as positive and area below counts as _____.

Example 3

Evaluate $\int_0^5 (x - 2)\,dx$ using geometry.

Integrable: if f is _____ on [a, b], then the definite integral exists.

Example 4

Evaluate $\int_{-3}^{3} \sqrt{9 - x^2}\, dx$.

Part 3 · Partner practice

Partner 1

Evaluate $\int_{0}^{4} (x + 1)\, dx$ using geometry.

Partner 2

(AP® Practice) For $f(x) = 2 - 3x$ on $[0, 4]$ with four subintervals of width 1, the right Riemann sum equals

(A) −22 (B) −24 (C) −28 (D) −10